

ICT702 Data Visualisation

Exploring Data Visually

Module 6



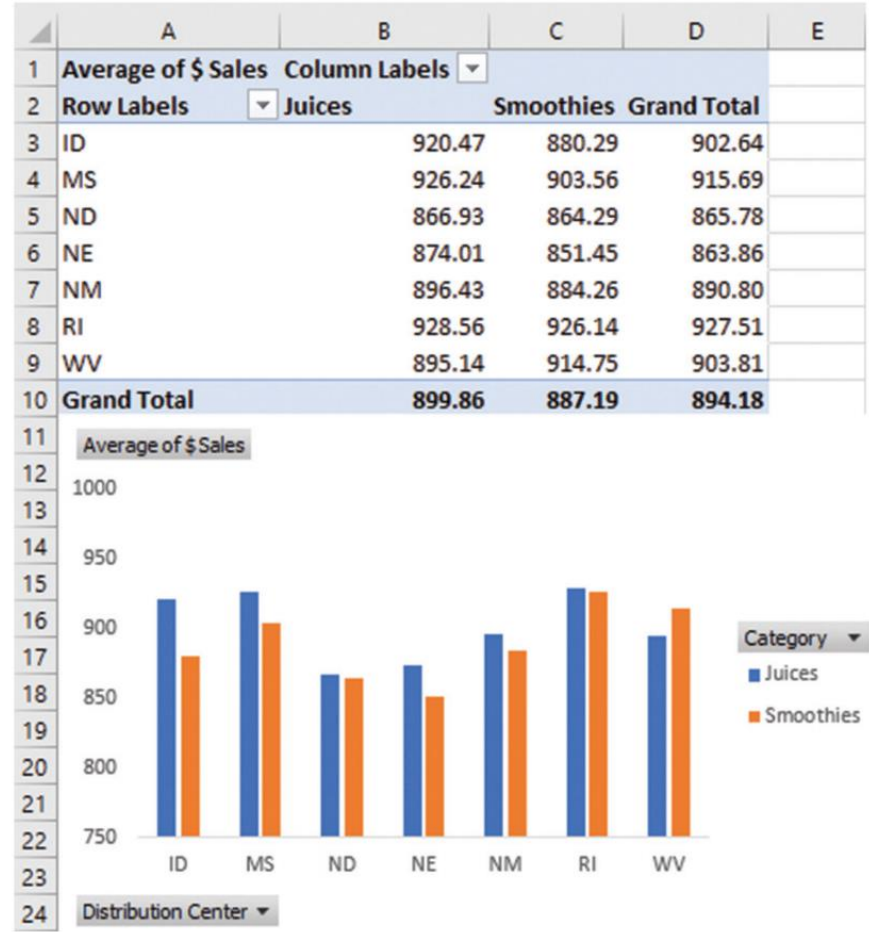
I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



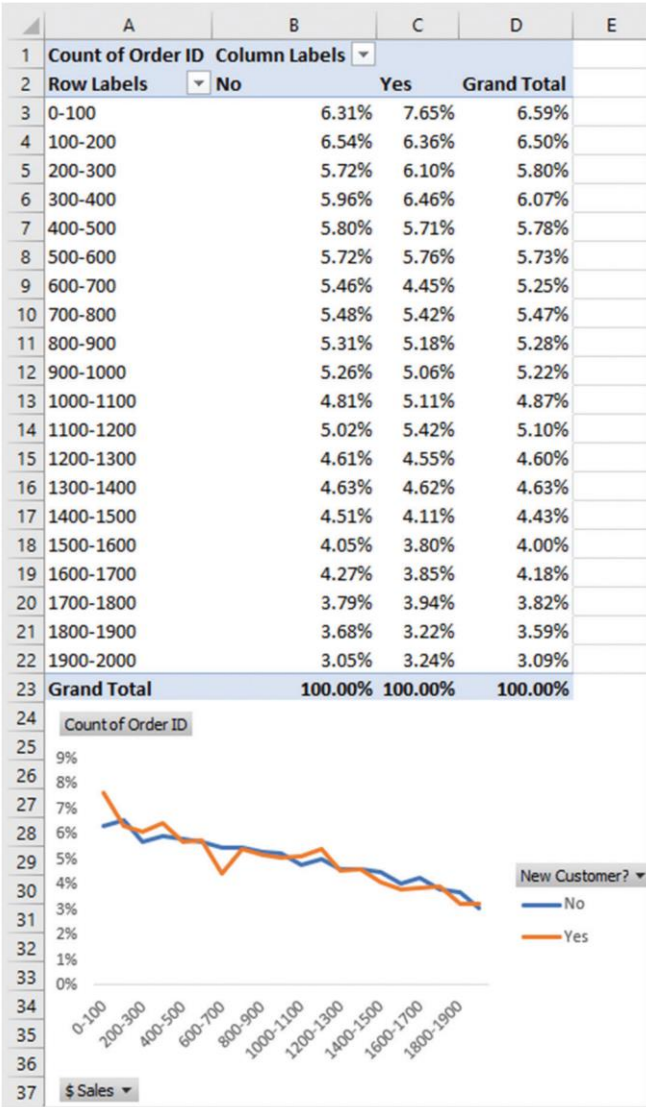
Part 2

EJB data: Crosstabulation

Categories: Crosstabulation of average \$ Sales over *Distribution Center* and *Category*



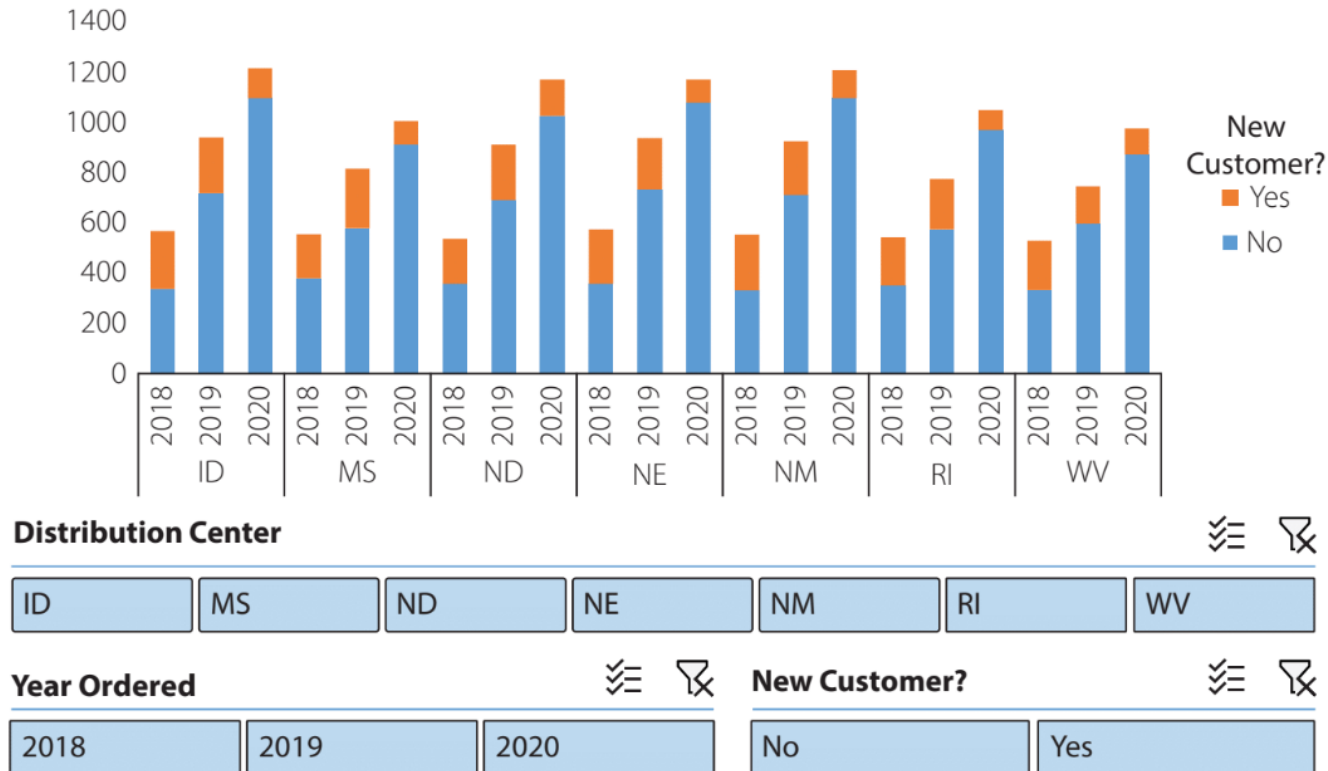
Grouped Values:
Crosstabulation
of Counting
Order ID over \$
Sales and *New
Customer*



Crosstabulation with Slicers

PivotChart for total sales with slicers for *Distribution Center*, *Year Ordered*, and *New Customer*

Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer



Application of Crosstabulation with Slicers

PivotChart with slicers for total *sales* to new customers shipped from New Mexico and Rhode Island in 2019 and 2020

Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer



This PivotChart allows the user to graphically display total dollar sales for any combination of years ordered, DCs, and new/existing customer status.

Association Between Two Quantitative Variables

A portion of New York City sub-borough data from the *NYC* file

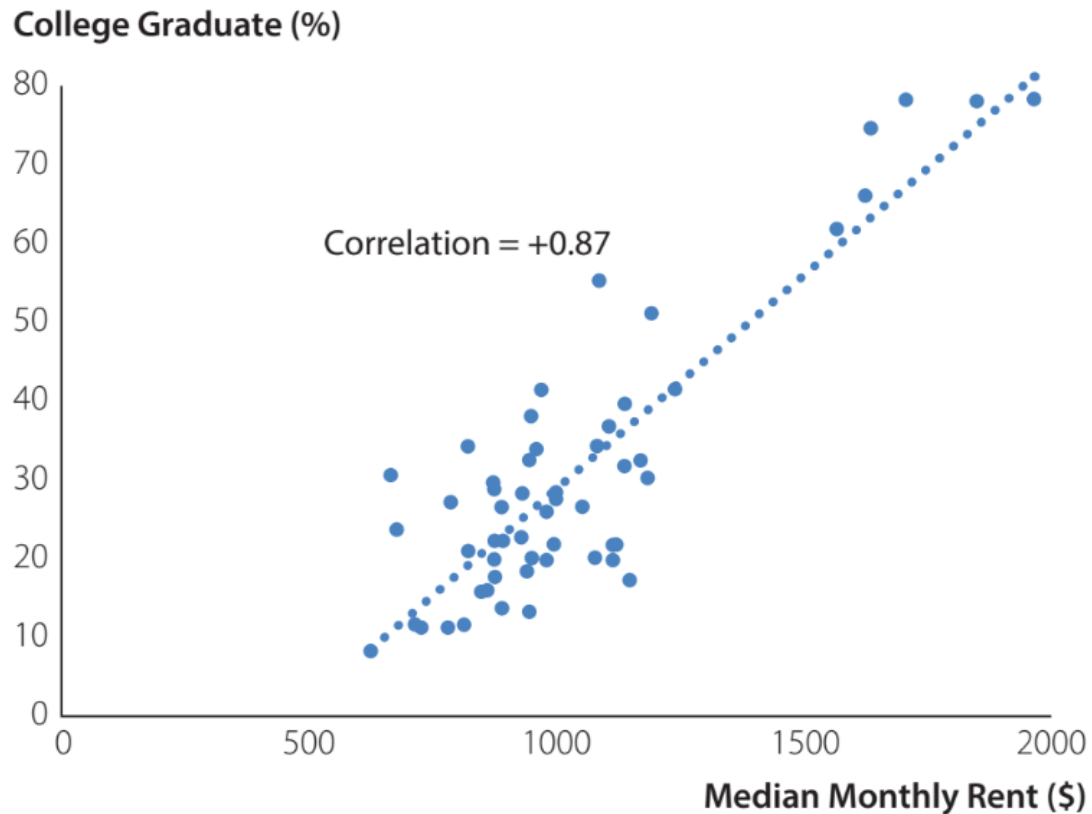
	A	B	C	D	E	F
1	Area (Sub-Borough)	Median Monthly Rent (\$)	Percentage College Graduates (%)	Poverty Rate (%)	Commute Time (min)	Borough
2	Astoria	1106	36.8	15.9	35.4	Queens
3	Bay Ridge	1082	34.3	15.6	41.9	Brooklyn
4	Bayside/Little Neck	1243	41.3	7.6	40.6	Queens
5	Bedford Stuyvesant	822	21.0	34.2	40.5	Brooklyn
6	Bensonhurst	876	17.7	14.4	44.0	Brooklyn

Correlation is a statistical measure of the strength of the linear relationship between two variables.

Positive Linear Association Between Variables

Scatter chart of *Percentage College Graduates* versus *Median Monthly Rent* with a linear trendline

Positive Relationship between Rent and Percentage College Graduates in a Sub-borough

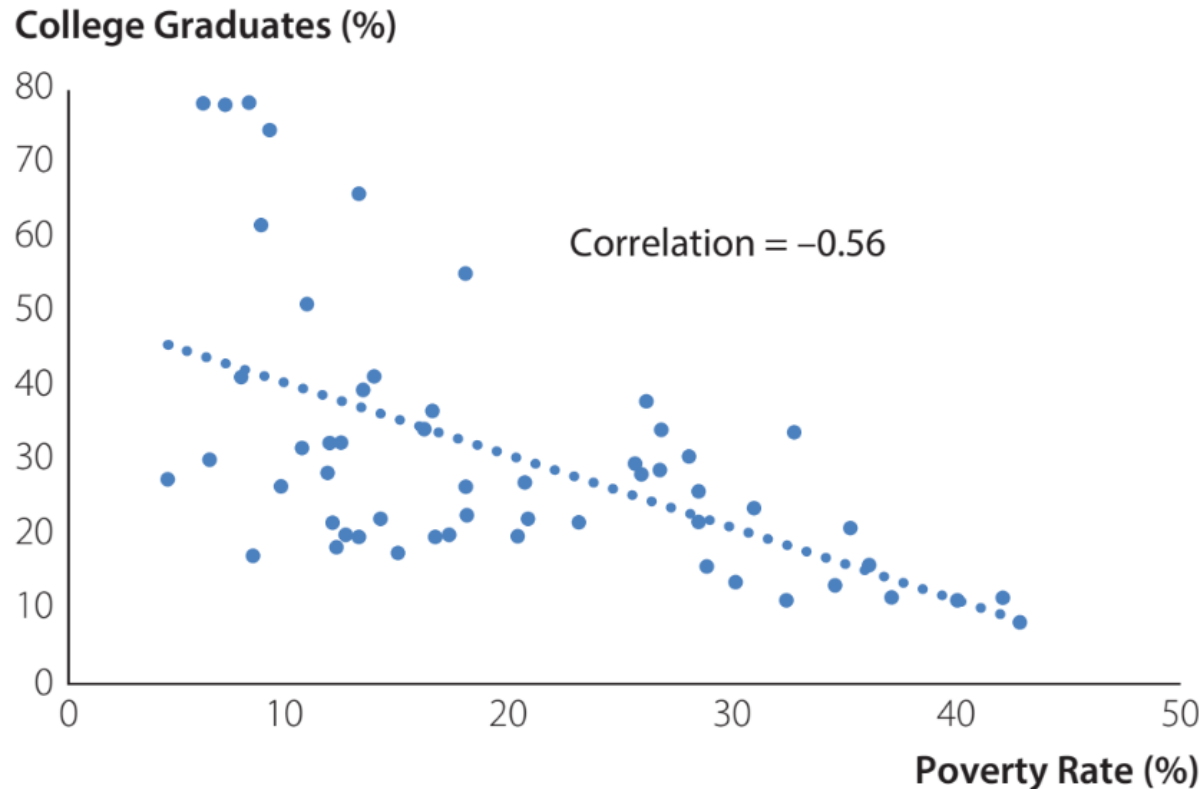


The strength of a correlation can be visually gauged by how close the data points cluster around the linear trendline.

Negative Linear Association Between Variables

Scatter chart of *Percentage College Graduates* versus *Poverty Rate* with a linear trendline

Negative Relationship between Poverty and Percentage College Graduates in a Sub-borough



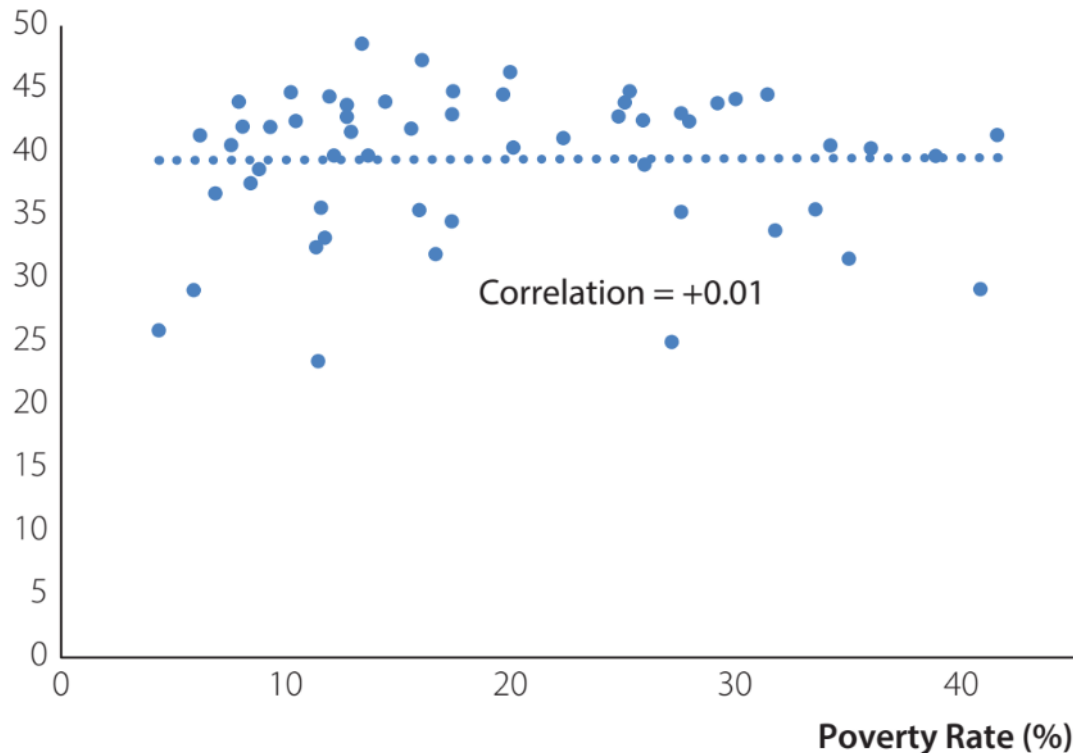
As the poverty rate increases in a sub-borough, the percentage of the sub-borough's inhabitants that are college graduates decreases.

Missing Association Between Variables

Scatter chart of Mean Commute Time versus Poverty Rate with a linear trendline

No Relationship between Poverty and Commute Time in a Sub-borough

Commute Time (Minutes)



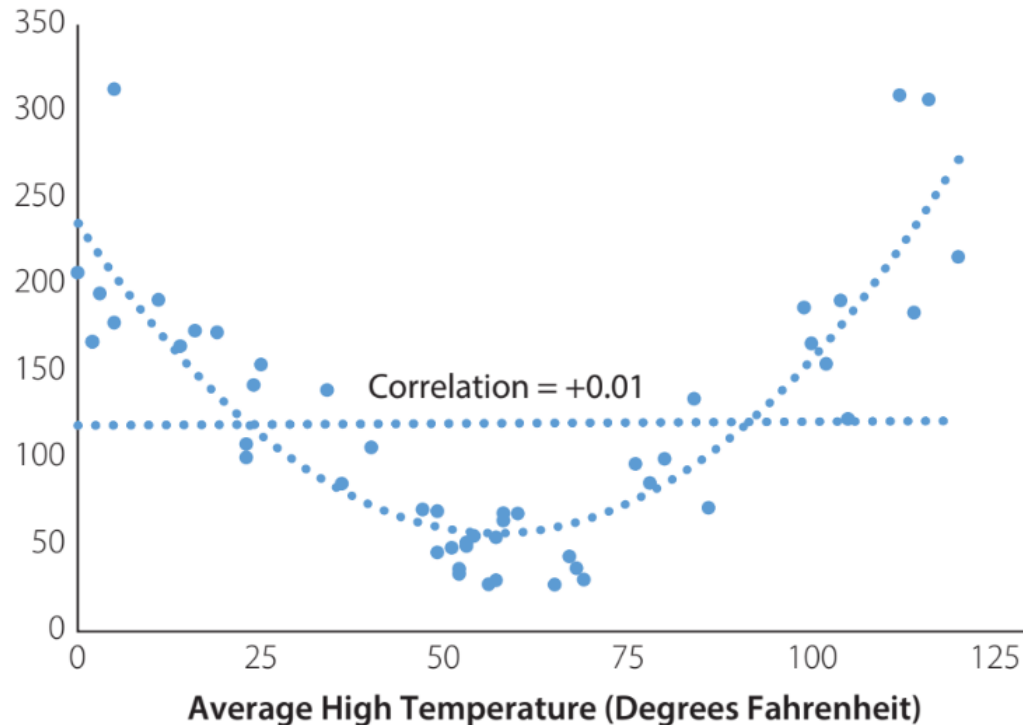
The data points poorly fit the linear trendline and the correlation is near zero.

Nonlinear Association Between Variables

Example of Nonlinear Relationship Producing a Near Zero Correlation

Nonlinear Relationship between Temperature and Heating/Cooling Bill

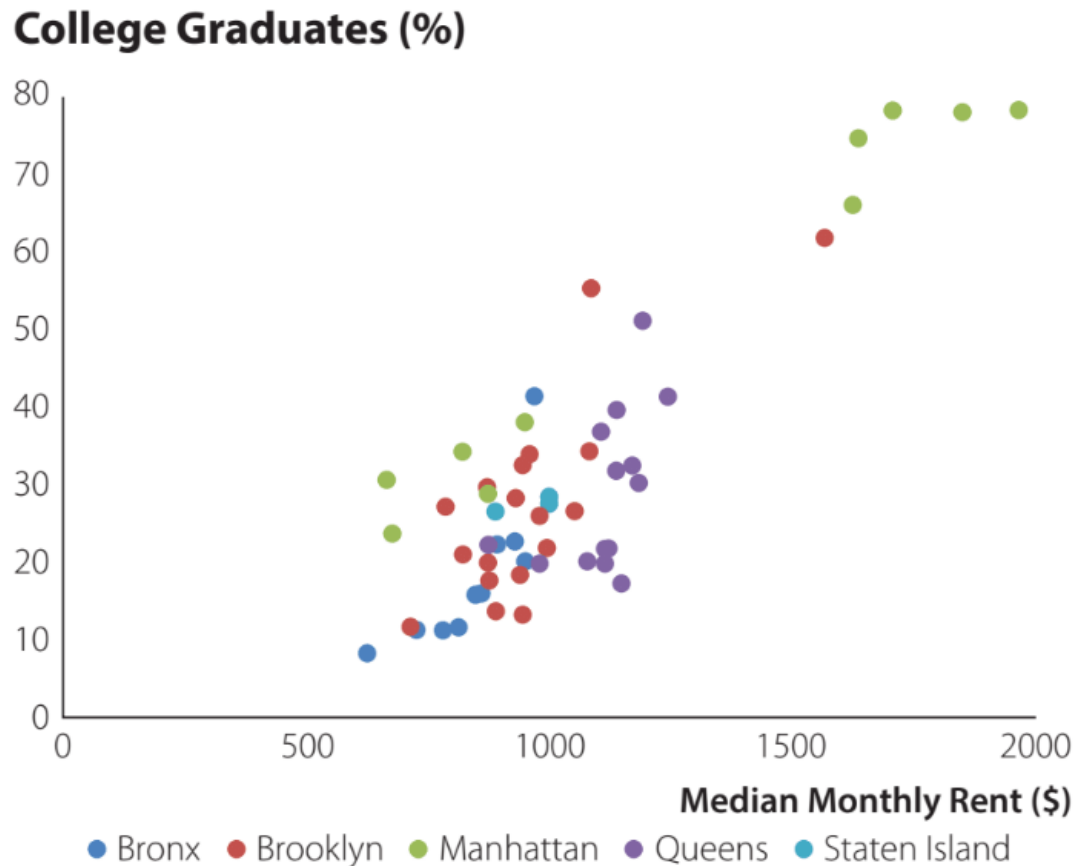
Monthly Heating/Cooling Bill (\$)



The nonlinear trendline outlines strong visual evidence of a nonlinear relationship between the two variables.

Association in Presence of a Categorical Variable

Using colour in a scatter chart to communicate a categorical variable

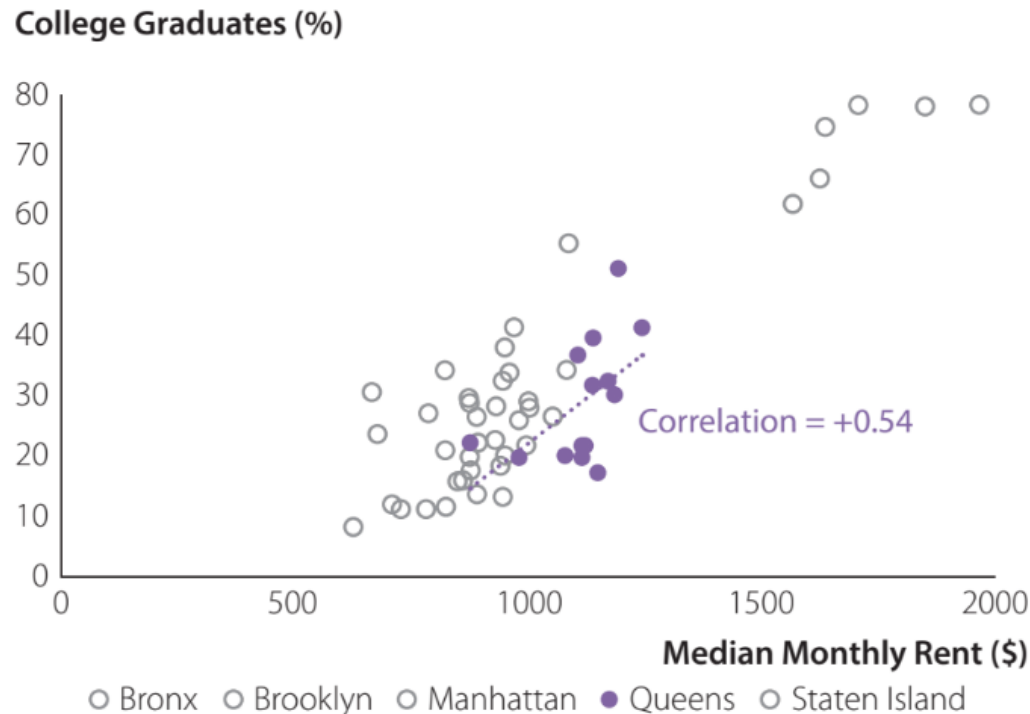


Using colour, we can distinguish the 55 observations (sub-boroughs) by their borough.

Highlighting Specific Value of a Categorical Variable

Using colour to highlight a specific borough (value of categorical variable)

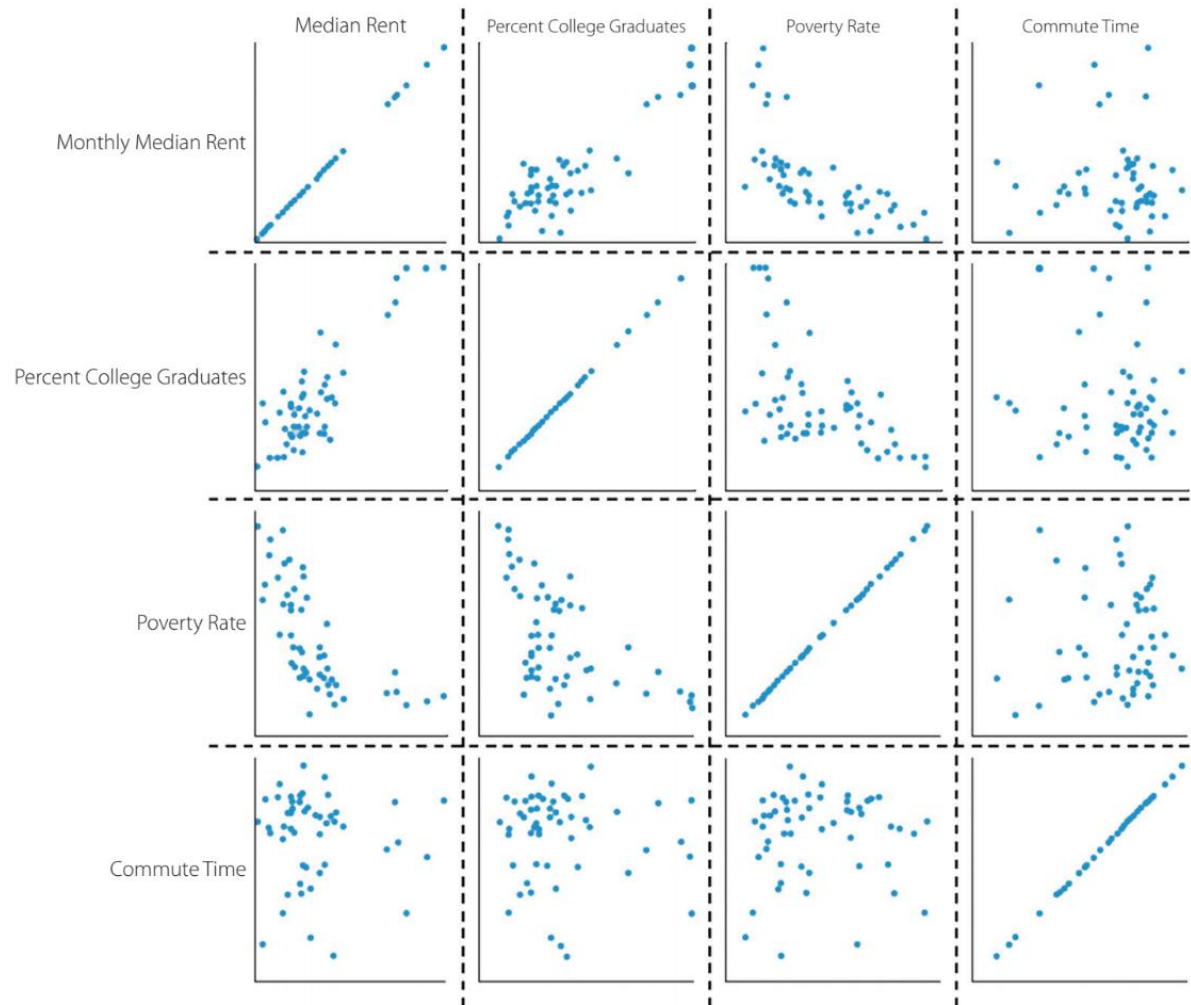
Correlation between Rent and Percentage College Graduates in Queens Weaker than in Other Sub-boroughs



The use of colour highlights how the strength of the correlation between monthly median rent and the percentage of college graduates is not as strong in Queens as it is over the other boroughs.

Association Between Multiple Variable Pairs

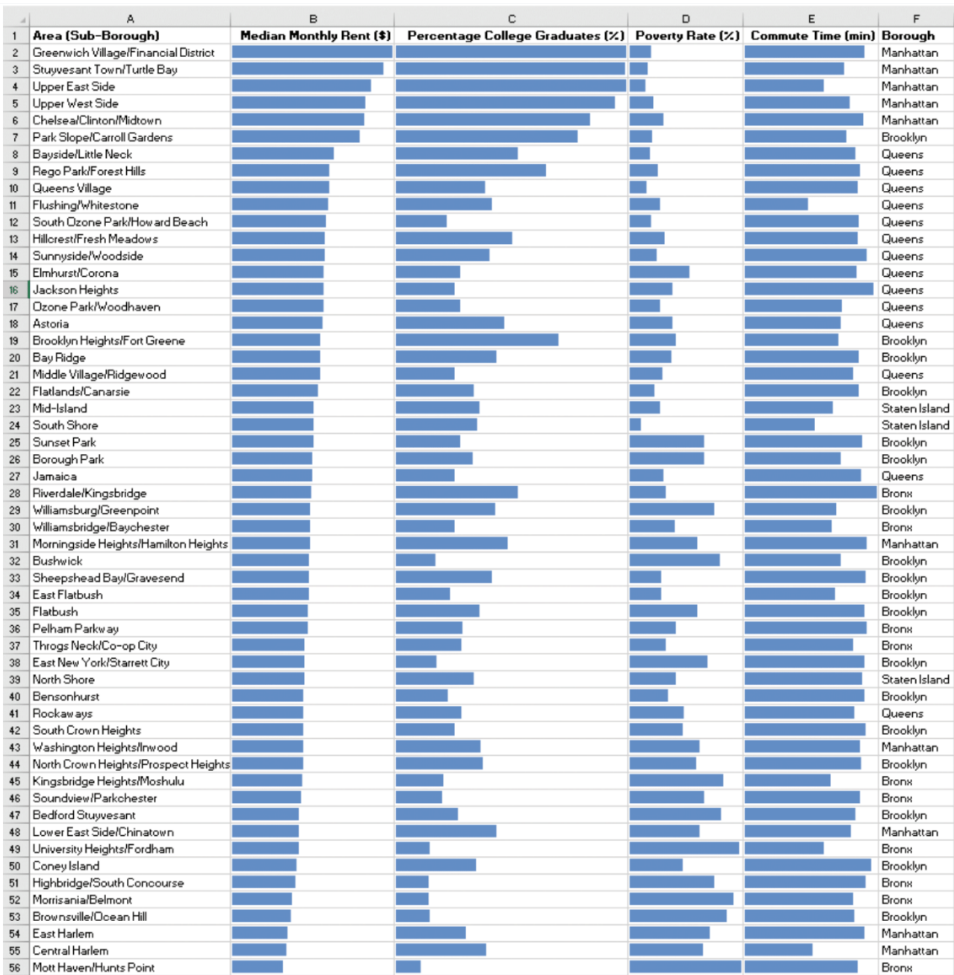
Scatter-chart matrix for New York City sub-borough data



A **scatter-chart matrix** visualises the relationship between each possible pair of quantitative variables in a data set.

Alternative Visualization for Multiple Associations

Table lens for New York City data



A **table lens** is an alternative way to a scatter-chart matrix to visualise relationships between different pairs of variables.



End of Part 2

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